

DINH CONG THANH

Senior AI Workflow Engineer | Full-stack Developer | AI Automation Engineer | Production RAG & Orchestration

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Showcase URL: <https://github.com/ThanhDC-FSD/Job-Ops-Console/tree/showcase>

Professional Summary

- Production-oriented engineer focused on *AI workflow orchestration*, reliable automation, and evidence-grounded outputs (RAG + validation) on CPU-friendly stacks.
- Designed and operationalized end-to-end pipelines: ingestion/normalization, retrieval + rerank, quality gates, deterministic fallbacks, and artifact rendering (TXT/DOCX/PDF) with auditable metadata.
- Strong delivery in enterprise environments: CI/CD automation, migration tooling, secure APIs, and resilient integrations with measurable efficiency gains.

Work Experience

Senior Full Stack Developer | Bosch Global Software Vietnam

Mar 2022 - Present

- Orchestrated an HP ALM → GitHub migration and delivery automation pipeline (Java, Maven, RTC integrations, Jenkins), preserving history/metadata and reducing manual release steps by ~60% while improving traceability.
- Automated CI/CD workflows (Jenkins Groovy, Docker, docker-compose, Nginx) with rollback patterns, cutting deployment turnaround time by ~30% and reducing repeatable release errors.
- Implemented resilient data workflows for PostgreSQL (SQLAlchemy/Alembic), including daily automated backups (pg_dump + restore validation) to improve recovery confidence and operational reliability.
- Productionized secure API services (Python FastAPI, SQLAlchemy) with enterprise authentication (Azure MSAL, JWT, cryptography), enabling controlled access to internal systems.

- Operationalized real-time workflow feedback using WebSockets and structured event logging, reducing time-to-diagnose for runtime issues by **~25%** during active development cycles.
- Built and hardened automation assistants: prompt template standardization (**React** + **TypeScript** + **Zod**), grammar **enhancement** (**LanguageTool** + **spaCy**), and tool integrations that **improved** consistency and reduced rework.
- Delivered RAG-enabled internal tooling (**FastAPI**, **LangChain**, **Chroma**) with practical retrieval + recommendation workflows to surface actionable knowledge and reduce time spent searching documentation.
- Built modern knowledge-assisted workflow tooling with **React**, **FastAPI**, MCP patterns, **LangChain**, and **Chroma** to combine project context, retrieval, and proactive risk guidance in a single operator flow, reducing issue-triage and context-gathering time by **~35%**.
- Built an LLM-assisted slide-generation workflow with **Python**, **Mermaid**, and an **Angular**-based UI to turn prompts and structured inputs into presentation-ready outlines, diagrams, and editable content, reducing manual first-draft slide preparation time by **~70%** while **improving** output consistency across iterations.

Personal Project | Render CV Automation and MP4 Processing & Translation Platform

- Orchestrated an end-to-end **AI automation** platform for job applications: LinkedIn crawling, normalization, triage queues, CV/JD fit evaluation, and assisted apply workflows in a single operational dashboard.
- Implemented a production-style **RAG** pipeline on **SQLite**: embedding + hybrid retrieval (semantic/keyword/metadata), reranking, evidence mapping, and prompt contracts to keep outputs grounded and auditable.
- Added reliability gates and deterministic fallbacks (no-op rewrite detection, coverage deltas, constraint filters, mandatory cover-letter enforcement), **improving** stability on CPU-only hardware and reducing failed runs by **~40%**.
- Automated document workflows to render tagged plain text into consistent **DOCX/PDF** artifacts (**python-docx**, **docx2pdf**), reducing manual formatting time by **~80%** and **improving** consistency across versions.
- Built automation pipelines for media processing: subtitle extraction, multilingual translation, and speech-to-text (**faster-whisper**, **Transformers**, **Google Translate API**), **improving** turnaround time for content localization by **~50%**.

- Delivered operational APIs and dashboards ([FastAPI](#), [Uvicorn](#), [Pydantic](#), [React](#), [Vite](#)) with analytics and stateful UX ([D3 Geo](#), [TopoJSON](#)) to support high-throughput review of large job datasets.
- Implemented a pragmatic local CI/CD loop (scheduled deployments, sanitized publisher repo, local bare Git remote) to continuously ship updates while isolating runtime secrets.

Senior Java Developer and Oracle Middleware Engineer | GTEL ICT

Sep 2020 - Feb 2022

- Directed backend delivery for a national-scale system ([Spring Boot](#), [Spring Security](#)), [improving](#) service robustness, access control, and operational stability.
- Implemented BI and [ETL](#) workflows ([Oracle OBIEE](#), [ODI](#)), enabling reliable analytics pipelines and maintainable fact/dimension models for reporting.
- Optimized integrations with [SQL Server](#) using caching ([EhCache](#)) and schedulers, reducing latency and [improving](#) steady-state performance for batch workloads.

Java Developer | FPT Software

Jun 2019 - Sep 2020

- Implemented data ingestion automation for [Oracle/PostgreSQL](#) APIs, hardening large Excel imports with validation and error taxonomies to reduce operational incidents.
- Automated repeatable test/data workflows ([Winium](#)), [improving](#) throughput and reducing manual QA effort for regression tasks.

Java Developer (Contractor) | Tien Phong Bank

Feb 2019 - Jun 2019

- Automated a Data Exchange BOT ([Java](#) Swing, [Selenium](#)) for scheduling, reporting (PDF/TXT), and database synchronization, reducing manual operational steps by [~70%](#).
- Implemented resilient ESB components ([ESQL](#)) to [improve](#) stability of card-service integrations and reduce recurring processing failures.

Java Developer | CODEGYM Vietnam

Aug 2018 - Feb 2019

- Delivered backend services (Spring Boot) and REST APIs with maintainable data models, improving delivery consistency across multiple training products.
- Implemented MySQL-backed persistence with unit tests and code review practices to improve correctness and reduce regressions.

Technical Skills

- **AI Workflow / RAG:** RAG pipelines, hybrid retrieval, reranking, evidence mapping, prompt contracts, validation gates, deterministic fallbacks, LangChain, Chroma, FAISS, sentence-transformers, qwen2.5 (offline gateway)
- **Workflow Automation:** Orchestration patterns, schedulers (APScheduler/cron), job queues, browser automation (Playwright), data normalization, fit scoring, integration reliability
- **Languages:** Java, Python, JavaScript (ES6+), TypeScript, SQL
- **Backend:** FastAPI, Starlette, SQLAlchemy, Pydantic, Uvicorn, Flask, Django, Spring Boot
- **Data / ETL:** pandas, numpy, openpyxl, XlsxWriter, ExcelJS, ETL pipelines, Oracle ODI/OBIEE
- **DevOps / Reliability:** Jenkins (Groovy), Docker, docker-compose, Nginx, Linux, rollback strategies, backup/restore validation
- **Databases:** PostgreSQL, SQLite, Oracle, SQL Server, MySQL
- **Frontend (Supportive):** React, Vite, HTML5, CSS3, Angular, D3 Geo, TopoJSON, world-atlas
- **Security:** Azure MSAL, JWT, passlib, cryptography, Spring Security
- **Others:** httpx, requests, Jinja2, python-docx, docx2pdf, python-pptx

Core Competencies

- AI Workflow Orchestration and Automation Engineering
- Production RAG Pipelines and Evidence-Grounded Generation
- Validation Workflows, Reliability Gates, and Deterministic Fallbacks
- CI/CD Automation, Migration Tooling, and Operational Hardening
- ETL/Data Workflows, Analytics, and High-Throughput Processing
- Secure API Design and Enterprise Authentication
- Cross-Functional Delivery, Ownership, and Collaboration